Experiment No 4

Title: Implementing FST as morphological (word) parser for noun inflection.

Program:

```
regular_nouns = {
    "bureau": "BUREAU",
    "bureaus": "BUREAU",

     "life": "LIFE",
    "lives": "LIFE",

    "piano": "PIANO",
    "pianos": "PIANO",

    "echo": "ECHO",
    "echoes": "ECHO",

    "zero": "ZERO",
    "zeros": "ZERO",

    "studio": "STUDIO",
    "studios": "STUDIO",

    "zoo": "ZOO",
    "zoos": "ZOO",

    "bamboo": "BAMBOO",
    "bamboos": "BAMBOO",

    "city": "CITY",
    "cities": "CITY",

    "country": "COUNTRY",
    "countries": "COUNTRY",

    "church": "CHURCH",
    "churches": "CHURCH",

    "knife": "KNIFE",
    "knives": "KNIFE"
}


irregular_nouns ={

    "analysis": "ANALYSIS",
    "analyses": "ANALYSIS",


    "criterion": "CRITERION",
```

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    "criteria": "CRITERION",

    "phenomenon": "PHENOMENON",
    "phenomena": "PHENOMENON",

    "diagnosis": "DIAGNOSIS",
    "diagnoses": "DIAGNOSIS",

    "men": "MAN",
    "man": "MAN",

    "women": "WOMAN",
    "woman": "WOMAN",

    "children": "CHILD",
    "child": "CHILD",

    "nucleus": "NUCLEUS",
    "nuclei": "NUCLEUS",

    "feet": "FOOT",
    "foot": "FOOT",

    "people": "PERSON",
    "person": "PERSON",

    "oxen": "OX",
    "ox": "OX",

    "fungus": "FUNGUS",
    "fungi": "FUNGUS"
}
def fst_parser(word):
  word = word.lower()
  # regular nouns
  if word in regular_nouns:
    root= regular_nouns[word]
    # plural ending in s
    if word.endswith('s') and not word.endswith('es'):
      intermediate=root + " ^ S #"
      number = "PL"

    # plural ending in es
```

```
    elif word.endswith('es') and not word.endswith('ies'):
      intermediate=root + " ^ ES #"
      number = "PL"

    # plural ending in ies
    elif word.endswith('ies'):
      intermediate=root + " ^ IES #"
      number = "PL"

    # singular
    else:
      intermediate = root + " #"
      number = "SG"

  # irreglar nouns
  elif word in irregular_nouns:
    root =irregular_nouns[word]
    # singular
    if word == root.lower():
      intermediate = root + " #"
      number = "SG"

    # plural
    else:
      intermediate = root + " #"
      number ="PL"

  # Not found
  else:
    print("Word not found in lexicon")
    return
  output =" "+ root + " + N + "+number
  print("INPUT: ",word.upper())
  print("INTERMEDIATE: ",intermediate)
  print("OUTPUT: ",output)

print("*"*50)
print(" FST MORPHOLOGICAL PARSER FOR NOUN INFLECTION")
print("*"*50)
while(True):
  word = input("Enter word or type exit to stop: ")
  if word.lower() == "exit":
    break
```

```
fst_parser(word)
print("-"*50)
```

OUTPUT

```
**************************************************
 FST MORPHOLOGICAL PARSER FOR NOUN INFLECTION
**************************************************
Enter word or type exit to stop: studios
INPUT:  STUDIOS
INTERMEDIATE:  STUDIO ^ S #
OUTPUT:   STUDIO + N + PL
--------------------------------------------------
Enter word or type exit to stop: knives
INPUT:  KNIVES
INTERMEDIATE:  KNIFE ^ ES #
OUTPUT:   KNIFE + N + PL
--------------------------------------------------
Enter word or type exit to stop: women
INPUT:  WOMEN
INTERMEDIATE:  WOMAN #
OUTPUT:   WOMAN + N + PL
--------------------------------------------------
Enter word or type exit to stop: criterion
INPUT:  CRITERION
INTERMEDIATE:  CRITERION #
OUTPUT:   CRITERION + N + SG
--------------------------------------------------
Enter word or type exit to stop: lice
Word not found in lexicon
--------------------------------------------------
Enter word or type exit to stop: exit
```